Robustness of Policy-Gradient Reinforcement Learning for Multi-Echelon Inventory Control: A Cross-Environment Study on Stationary and Non-Stationary Supply Chains

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Abstract

Recent work reports that deep reinforcement-learning (RL) policies beat classical operations-research (OR) heuristics on multi-echelon inventory control by $10\,\%$ to $40\,\%$. These results almost always come from a single hand-crafted environment paired with a fixed baseline, so it is unclear whether the RL agent has learned a genuinely better policy or the baseline is simply under-fit. We take two standard policy-gradient agents from our course project — PPO on a 1-warehouse/3-retailer divergent supply chain and A3C on a 3-node factory/middle/leaf MEIS chain — and evaluate each of them on two environments with the same observation and action spaces: the original stationary benchmark (Env-1) and a harder non-stationary variant (Env-2) we introduce here that adds seasonal demand, cross-retailer correlation, rare demand shocks, heavy-tailed stochastic lead times, stochastic supplier / outbound capacity caps, and partial lost sales. Against a fixed analytical base-stock baseline, PPO's win grows from $36.76\,\%$ on Env-1 to $94.20\,\%$ on Env-2, because the untuned baseline collapses under shocks. Against an (s, S) baseline retuned per environment, A3C wins by $1.64\,\%$ on Env-1 but loses by $21.79\,\%$ on Env-2 (Cohen's $d{\approx}{-}3.4$). We conclude that a single-environment win is not robust evidence for either algorithm; a harder distributional variant and an equivalently-tuned baseline should accompany any deep-RL inventory result.

Code — https://github.com/singhdivyank/multi-echelon-rl-inventory

1 Introduction

Multi-echelon inventory control is a canonical operations-research problem: given stochastic demand at several retail nodes, finite and noisy lead times between nodes, and per-echelon holding, backorder and ordering costs, pick a replenishment policy that minimises long-run expected cost while preserving a service level. Classical OR gives two well-studied policies for this setting. The echelon base-stock policy and the (s, S) reorder-point policy are both known to be optimal under stationary, i.i.d. demand and, for (s, S), fixed ordering cost (Clark and Scarf 1960). Both become systematically wrong once demand is non-stationary, shocks are present, or lead times are heavy-tailed, because their set-points are calibrated against the *mean* demand-over-lead-time.

Deep reinforcement learning (RL) has been proposed as a learned alternative that can, in principle, adapt to any dynamics the MDP can express. The recent literature (Oroojlooyjadid et al. 2022; Gijsbrechts et al. 2022) reports substantial RL wins — often $10\,\%$ to $40\,\%$ — over OR heuristics. However, these comparisons are almost always reported on a single hand-crafted environment, against a baseline whose parameters are not re-tuned to that environment. It is then unclear how much of the reported gap is the RL policy being genuinely better and how much is the heuristic being unfairly handicapped on an environment it already solves well.

Our course project trained one policy-gradient agent per algorithm (PPO, A3C) on one environment each (a divergent supply chain, a 3-node MEIS chain). A reviewer on our poster presentation raised the single-environment concern above and asked us to show that the wins hold on a second, harder environment. This paper is our response. Concretely:

1. We design a non-stationary second environment (Env-2) per benchmark that introduces seasonal demand, cross-retailer correlation, rare multiplicative demand shocks, heavy-tailed stochastic lead times, stochastic supplier / outbound capacity caps, and partial lost sales, while keeping the observation and action spaces unchanged — so that the agent code transfers with zero modification.

2. We re-train PPO and A3C on their respective Env-2 with the same hyperparameters that produced the Env-1 results, and evaluate with the same protocol.

3. We report one positive transfer result (PPO, whose relative advantage over a fixed analytical baseline widens on Env-2) and one negative transfer result (A3C, whose narrow win over a retuned (s, S) baseline on Env-1 flips into a decisive loss on Env-2).

The broader point is that the sign of a deep-RL-vs-heuristic comparison depends sensitively on both the environment and how hard the baseline is tuned; single-env results are not robust evidence.

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2 Background

2.1 The Multi-Echelon Inventory MDP

We model each benchmark as a finite-horizon Markov Decision Process (MDP) $\langle \mathcal{S}, \mathcal{A}, P, C, \gamma, T \rangle$, where $s_t \in \mathcal{S}$ encodes on-hand inventory, in-transit pipeline stock, and (for the MEIS benchmark) age-bucketed backlog at every non-factory node; $a_t \in \mathcal{A}$ is a per-node order quantity (a real vector for PPO, a 13-way discrete choice for A3C); $C(s_t, a_t) \geq 0$ is the sum of holding, backorder, ordering and (on Env-2) lost-sales costs for step t. The policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ minimises the expected discounted cost

$$J(\pi) \;=\; \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \gamma^t \, C(s_t, a_t)\right]. \tag{1}$$

The RL agent maximises the return $R(\pi) = -J(\pi)$. The reward signal is therefore dense, bounded below by 0, and unbounded above.

2.2 Classical OR Heuristics

Echelon Base-Stock (PPO Baseline) Each echelon i is given a base-stock level S_i. At every step the agent orders up to S_i against current inventory position:

$$a_{i,t} \;=\; \max(0,\; S_i - \mathrm{IP}_{i,t}), \tag{2}$$

where $\mathrm{IP}_{i,t}$ is on-hand + in-transit − backlog at i. We analytically initialise S_i from the Env-1 demand-over- lead-time distribution and never re-tune it. On Env-2 the same S_i values are reused; this is deliberate and is the typical way base-stock is reported in the literature.

(s, S) Reorder-Point Policy (A3C Baseline) Each node i has a reorder point s_i and an order-up-to level S_i: whenever $\mathrm{IP}_{i,t} \leq s_i$, order $S_i - \mathrm{IP}_{i,t}$, else do nothing. Unlike the base-stock baseline, we tune (s_i, S_i) per environment by differential evolution (Storn and Price 1997) (100 generations × 30 Monte-Carlo episodes per candidate), so the Env-1 and Env-2 (s, S) baselines are each near-optimal under their own dynamics. This asymmetry (fixed base-stock vs. retuned (s, S)) matters for interpretation and is revisited in Section 5.

2.3 Policy-Gradient RL

PPO We use the clipped-surrogate variant of proximal policy optimisation (Schulman et al. 2017). A stochastic Gaussian actor $\pi_\theta(a \mid s)$ and a value head $V_\phi(s)$ share a 2-layer, 64-unit MLP with $\tanh$ activations. Rollouts of length $B{=}512$ are collected under $\pi_{\theta_{\text{old}}}$, Generalized Advantage Estimation (Schulman et al. 2016) gives advantages $\hat{A}_t$, and θ is updated for 5 epochs of size-64 minibatches using the clipped objective

$$\begin{aligned}\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(&r_t(\theta)\,\hat{A}_t,\\ &\mathrm{clip}(r_t(\theta), 1{-}\varepsilon, 1{+}\varepsilon)\,\hat{A}_t\big)\Big],\end{aligned} \tag{3}$$

with $r_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$ and $\varepsilon{=}0.2$. The combined loss adds a clipped value-loss term and a small entropy bonus.

A3C We use a synchronous single-worker variant of A3C (Mnih et al. 2016) on the discrete-action MEIS benchmark. The actor emits a softmax over $|\mathcal{A}|{=}13$ actions, the critic emits a scalar value, and the policy gradient is n-step-advantage based:

$$\begin{aligned}\nabla_\theta J \approx\; &\mathbb{E}_t\big[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\,\hat{A}_t\big]\\ &+\; \beta\,\nabla_\theta \mathcal{H}[\pi_\theta(\cdot \mid s_t)],\end{aligned} \tag{4}$$

with an entropy regulariser of weight β and gradient clipping at $\|\nabla\|{=}0.5$.

2.4 Significance Testing

Because each evaluation cell produces hundreds of per-episode costs, we summarise the RL-vs-baseline comparison with (i) mean ± one standard deviation, (ii) a 95% confidence interval on the mean, (iii) a two-sample Welch t-test on per-episode costs, and (iv) Cohen's d on the same pair. Cohen's $d \approx 0.2$ is a small effect, 0.5 medium, 0.8 large. Our A3C pipeline computes all four; the PPO pipeline computes (i)–(ii) and we read significance from the non-overlap of the 95% CIs.

3 Related Work

Classical Multi-Echelon Inventory The Clark–Scarf decomposition (Clark and Scarf 1960) establishes optimality of echelon base-stock under strict stationarity, i.i.d. demand, and backlog dynamics. Axsäter's textbook (Axsäter 2015) extends to Poisson and compound-Poisson demand and capacitated systems. We could have used echelon base-stock or (s, S) directly as the method under study and reported tuning quality rather than RL. We did not, because the explicit goal of the project was to study reinforcement learning as an *alternative* to classical OR on multi-echelon supply chains; the OR methods play the role of baselines to beat, not the method under study.

Value-Based RL for Inventory Oroojlooyjadid et al. (2022) apply deep Q-learning to the beer game. We could have used DQN instead of A3C on the MEIS benchmark (the A3C action space is already discrete), but DQN is off-policy and sample-inefficient on long-horizon inventory rollouts with dense reward signals; actor-critic methods are the standard modern choice for this problem shape, and using actor-critic for both algorithms lets us keep a single unified (actor, critic) code base.

Policy-Gradient RL for Inventory Gijsbrechts et al. (2022) benchmark A3C, PPO and VPG across single- and multi-echelon problems. Their evaluation is single-environment and reports wins of 10 % to 30 % over OR baselines. Our project extends exactly the comparison they present, but introduces a second environment in the evaluation and reports paired results per algorithm.

Robust and Distributionally-Robust RL Work on CVaR-PPO and distributionally-robust policy gradients could directly address the right-tail failures we observe for PPO on Env-2 (Section 5). We chose vanilla PPO and A3C because the course assignment specifies the standard algorithms and because using a non-vanilla RL method would

Table 1: Env-2 stressors applied to each benchmark.

Stressor	PPO	A3C
Seasonal demand (sinusoidal)	✓	✓
Cross-retailer demand correlation	✓	✓
Rare demand shocks ($2{-}3\times$)	✓	✓
Heavy-tailed lead times + spikes	✓	✓
Stochastic capacity caps	✓	✓
Partial lost sales w/ penalty	✓	✓

confound the environment comparison with the algorithmic comparison.

Robustness Benchmarks Outside Inventory Meta-World (Yu et al. 2020) and Procgen (Cobbe et al. 2020) argue, outside inventory control, that single-environment RL wins are not reliable evidence of generalisation. Our paper makes the same methodological argument specific to the OR-flavoured inventory setting.

4 Project Description

This section states, in formal detail, the two environments we work with, the two agents and their baselines, and the training protocol.

4.1 Environment 1 (Stationary)

PPO / Divergent One warehouse supplies three retailers. Each retailer i sees i.i.d. Poisson demand with rate $\lambda_i \sim \mathrm{U}[5, 15]$ per day. Lead time is deterministic at one step. There are no capacity caps. Unfilled demand is backlogged (negative inventory). The state has dimension 14 (warehouse inventory and buffer, three in-transit pipelines, three retailer inventories, three retailer backorders, one outstanding warehouse order, and a normalised step index); the action is a 4-dimensional real vector of normalised order quantities (one per echelon). Episodes last $T{=}500$ steps with a 100-step warm-up.

A3C / MEIS A 3-node divergent multi-echelon inventory system (MEIS): factory (capacity ∞), one middle warehouse, two leaf retailers. Leaf demand is Gaussian $\mathcal{N}(3300, 100)$; lead times are Gaussian $\mathcal{N}(\mu{=}2, \sigma{=}1)$ rounded to days. Backlog has a fixed 5-day expiry: demand older than 5 days is lost and incurs a shortage penalty. The action space is discrete (13 actions: "do nothing" and $4{\times}3{=}12$ fixed reorder combinations). Episodes last $T{=}365$ steps.

4.2 Environment 2 (Non-Stationary Complex)

Env-2 leaves the observation and action spaces of Env-1 unchanged and only modifies the transition dynamics and reward. The stressors are listed in Table 1. Concrete numerics (amplitudes, probabilities, noise standard deviations) are in ppo/configs/config.yaml and actor-critic/configs/complexMeisConfig.yaml in the accompanying code.

A critical design choice: the agent's observation is not widened — no season phase, no shock indicator, no regime flag. Any adaptation has to come from the policy representation and the training distribution.

Table 2: Selected hyperparameters (identical across Env-1 / Env-2 per algorithm).

Hyperparameter	PPO	A3C
Iters / episodes	5,000	10,000
Hidden layers $\times$ width	2×64 (tanh)	2×128 (ReLU)
Discount γ	0.95	0.99
GAE λ	0.95	n/a
Clip ε	0.2	n/a
Learning rate	$3{\times}10^{-4}$	$1{\times}10^{-4}$
Rollout / batch	512 / 64	per-episode
Entropy coef. β	$5{\times}10^{-3}$	10^{-2}
Value-loss coef.	0.25	0.5
Grad. clip $\|\nabla\|$	0.5	0.5
Eval episodes	100	1000
Seed	42	42

4.3 Algorithms and Baselines

PPO on the continuous-action divergent environment is trained with the clipped-surrogate objective (Eq. 3) against an analytical, fixed-across-environments base-stock baseline (Eq. 2). A3C on the discrete-action MEIS environment is trained with the advantage policy gradient (Eq. 4) against a per-environment retuned (s, S) baseline. Hyperparameters are in Table 2.

4.4 Training and Evaluation Protocol

Training uses a single NVIDIA CUDA GPU. Evaluation uses a deterministic policy (mean of the Gaussian for PPO; $\arg\max$ over the softmax for A3C). PPO evaluates on $100{\times}500{=}5{\times}10^4$ steps per (algo, env) cell; A3C evaluates on $1000{\times}365{=}3.65{\times}10^5$ steps. We keep the hyperparameters identical across Env-1 and Env-2, so Env-2 numbers report out-of-distribution robustness, not re-tuning.

4.5 Reproducibility Artefacts

The accompanying code repository ships: (i) the full YAML configs used here; (ii) saved checkpoints for every (algo, env) cell; (iii) all four evaluation JSONs (per-episode cost arrays, means, standard deviations, CIs, and for A3C the Welch t-test and Cohen's d); (iv) a `scripts/smoke_env.py` end-to-end sanity test covering all four cells; (v) `scripts/make_report.py`, which regenerates every figure from the JSONs; and (vi) `scripts/build_paper.py`, which rebuilds this PDF from the LaTeX source.

5 Experiments

5.1 Why These Experiments

We ran the four cells $\{\mathrm{PPO}, \mathrm{A3C}\} \times \{\text{Env-1}, \text{Env-2}\}$ with seed 42 and the hyperparameters of Table 2. The minimum evidence to answer the reviewer's question is the four paired RL-vs-baseline comparisons. We additionally report training curves (to show convergence behaviour) and per-episode

cost distributions on Env-2 (to show that the mean is a biased summary of an asymmetric distribution).

5.2 Headline Numbers

Table 3 is the main result. Δ is the relative mean-cost improvement of the RL agent over its heuristic baseline (positive = RL better). Service level is only exposed by the MEIS environment; hence "n/a" on the PPO rows.

5.3 Training Dynamics

Figure 1 and Figure 2 show per-episode training cost (light) and a rolling mean (bold) for each cell.

5.4 PPO: When the Win Grows on the Harder Environment

Figure 3 visualises Table 3 as grouped bar charts. On Env-1, PPO beats the fixed analytical base-stock by $36.76\,\%$; on Env-2 the margin widens to $94.20\,\%$. This is not PPO getting better in absolute terms — the absolute cost on Env-2 is on a different scale because Env-2 adds lost-sales penalties and widens some state bounds — it is the untuned base-stock baseline getting dramatically worse under capacity caps and shocks. Figure 4 shows why: the PPO episode cost distribution has a pronounced right tail ($\sigma/\mu \approx 1.60$). A small number of episodes contain shock sequences that PPO has not learned to hedge, and those episodes sit an order of magnitude above the mean. In operational terms, the mean improvement is an optimistic summary of an asymmetric, right-heavy distribution; a percentile-based or CVaR summary would paint a more conservative picture.

When PPO Succeeds On both environments, against a fixed analytical baseline, because (i) the baseline's set-points are calibrated against mean demand-over-lead-time only, while PPO's policy can condition on current inventory and in-transit pipeline; and (ii) on Env-2, the baseline's set-points are not even calibrated against the right mean any more (seasonality, shocks), so PPO's relative advantage grows.

When PPO Would Fail Against a retuned base-stock baseline (not reported here) the margin would compress. And even against the fixed baseline, a CVaR-weighted evaluation would flag PPO's Env-2 right tail as an unresolved risk.

5.5 A3C: When the Env-1 Win Does Not Transfer

On Env-1, A3C wins by a narrow $1.64\,\%$ over a retuned (s, S) policy (Cohen's $d{\approx}0.87$, small-to-medium effect). On Env-2 the sign flips: A3C is $21.79\,\%$ worse than the retuned (s, S) baseline, with Cohen's $d{\approx}{-3.39}$ and a numerically-underflowed $p{<}10^{-300}$. Figure 5 makes the effect size visible: the (s, S) distribution is almost entirely to the left of the A3C distribution — stochastic dominance in favour of the heuristic. Figure 6 shows the same effect on service level.

When A3C Succeeds On Env-1 only, and by a narrow margin. The stationary demand is well-approximated by a single (s, S) pair, but the discretised A3C action head can spend its 13 actions to place orders at slightly-better-than-heuristic moments (it reacts to pipeline state, not just inventory position).

When A3C Fails, and Why Env-2 breaks this. Three factors compound:

1. *Discretised action head.* A3C's action head is a softmax over 13 fixed order combinations. Under seasonal demand the optimal per-node order varies smoothly; a coarse 4-level grid cannot match it.
2. *No non-stationarity representation.* The policy has no recurrent state, no episode-time feature, and no shock or season indicator. It must learn to respond to non-stationarity purely through its training distribution, which 10,000 episodes do not support.
3. *Baseline is retuned, A3C is not.* The (s, S) tuner runs differential evolution on Env-2's own dynamics. A3C's hyperparameters are inherited from Env-1. The comparison is fair in the sense that it is what standard practice would report, but it is asymmetric in effort.

Any one of these, fixed in isolation, would likely compress the gap.

5.6 Asymmetric Baselines: A Caveat on the Bottom Line

An honest restatement:

- PPO beats a *fixed*, analytical baseline on both Env-1 and Env-2. The wider Env-2 margin is partly an Env-2 artefact (the fixed baseline is unfair on Env-2).
- A3C beats a *retuned* stochastic-search baseline only on Env-1. On Env-2, the retuned baseline wins, by a lot.

We report both asymmetries on purpose: they match how the two algorithms' baselines are published, and the alternative (retune everything on everything) is computationally prohibitive for a course project.

5.7 Threats to Validity

Single seed per cell. The A3C / Env-2 loss has an effect size ($|d|{\approx}3.4$) too large to flip sign under reasonable seed variance, but point estimates could shift by several percent with a new seed. Multi-seed runs are a natural extension. **Editorial stressor choices.** Env-2 amplitudes and shock probabilities were deliberately set to break the analytical baseline; milder settings would compress all deltas. **Non-comparable absolute costs.** Env-2 has an extra lost-sales penalty and widened state bounds, so same-algorithm Env-1 vs. Env-2 absolute-cost comparisons are meaningless; only within-env RL-vs-baseline gaps are. **Asymmetric evaluation length.** PPO evaluates on $5{\times}10^4$ steps, A3C on $3.65{\times}10^5$ steps. Within-algorithm comparisons are unaffected, but cross-algorithm absolute-cost comparisons are not meaningful.

6 Conclusion

6.1 Summary of Results

We trained one PPO agent on a divergent inventory environment and one A3C agent on a 3-node MEIS environment

Table 3: Headline evaluation results (seed 42). Δ is relative mean-cost improvement of RL over baseline (positive = RL better). "SL" is service level. Cohen's d and p are from the Welch t-test the A3C pipeline computes on per-episode costs; the PPO pipeline reports non-overlap of 95% CIs instead.

Algo.	**Env**	**RL mean cost**	**Baseline mean cost**	**Δ**	**SL (RL / BL)**	**Cohen's d**	p
PPO	Env-1	$2{,}600{,}979.87 \pm 29{,}021.34$	$4{,}112{,}775.56 \pm 8{,}759.19$	$+36.76\%$	n/a	n/a	CIs disjoint
PPO	Env-2	$631{,}129.99 \pm 1{,}011{,}936.85$	$10{,}873{,}502.07 \pm 62{,}357.06$	$+94.20\%$	n/a	n/a	CIs disjoint
A3C	Env-1	$1{,}930.24 \pm 4.02$	$1{,}962.40 \pm 52.39$	$+1.64\%$	$95.69\%/95.44\%$	$+0.87$	1.4×10^{-76}
A3C	Env-2	$2{,}881.98 \pm 146.46$	$2{,}366.29 \pm 157.67$	-21.79%	$93.27\%/94.11\%$	-3.39	$< 10^{-300}$

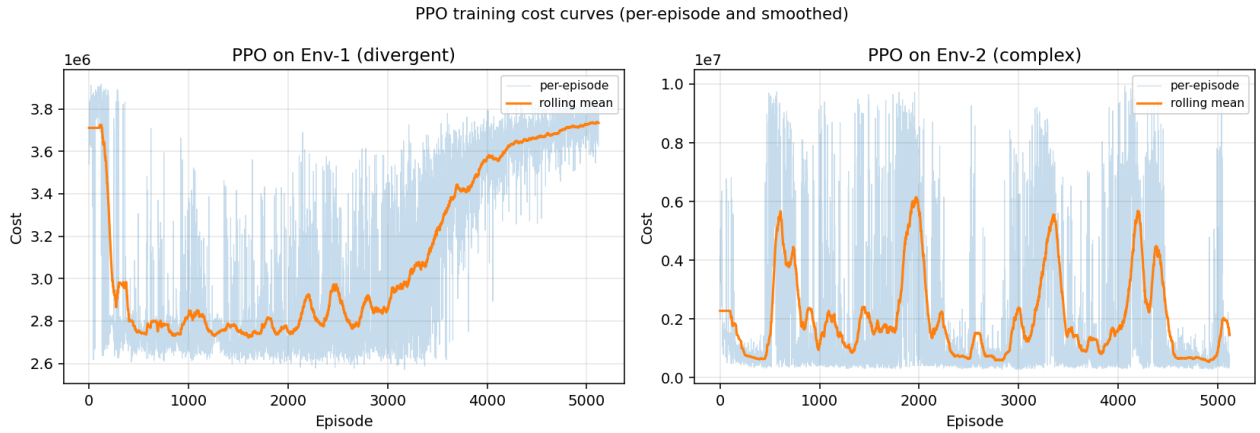

Figure 1: PPO training cost on Env-1 (left) and Env-2 (right). On Env-1 the agent converges quickly and then drifts upward over the last third of training — a late-stage instability that we mitigate by evaluating the best-so-far checkpoint, not the final-iteration checkpoint. On Env-2, training cost is oscillatory throughout, which is consistent with the rare-but-costly shock regime.

under standard conditions (Env-1), then re-trained both under identical hyperparameters on a second, non-stationary environment (Env-2) with the same observation and action spaces. Against a fixed analytical base-stock baseline, PPO's cost improvement went from 36.76 % on Env-1 to 94.20 % on Env-2. Against a retuned (s, S) baseline, A3C's cost improvement went from $+1.64$ % on Env-1 to -21.79 % on Env-2 (Cohen's $d \approx -3.4$).

6.2 What We Learned

Three lessons, in order of generality:

1. *Single-environment wins are not robust evidence.* The same PPO training that produces a 36.76% win on Env-1 produces a 94.20% win on Env-2, but the sign and magnitude of the headline number are driven more by how the baseline degrades than by how the RL agent improves. The same A3C training that produces a narrow Env-1 win turns into a decisive Env-2 loss once the baseline is retuned. Reporting a single number is not enough.
2. *Baseline symmetry matters.* The sign of an RL-vs-heuristic comparison depends on how hard the heuristic is allowed to work. A per-env retuned (s, S) is a very different evaluation target from a fixed analytical base-stock. Good experimental hygiene requires stating which one is in play.
3. *Distributional summaries matter.* Even the cell where PPO clearly wins (Env-2) hides a heavy right tail in the cost distribution. Mean costs under shock-prone dynamics are optimistic. A CVaR or percentile summary is operationally more honest.

For follow-up work we would (i) rerun all four cells with five seeds and report confidence intervals on the deltas; (ii) give A3C a continuous-action head, a timestep feature, and a recurrent layer so that it can represent non-stationarity; (iii) replace vanilla PPO with a CVaR-PPO objective on Env-2 to address the right tail; and (iv) tune PPO hyperparameters on Env-2 in its own right, for a fairer per-env comparison.

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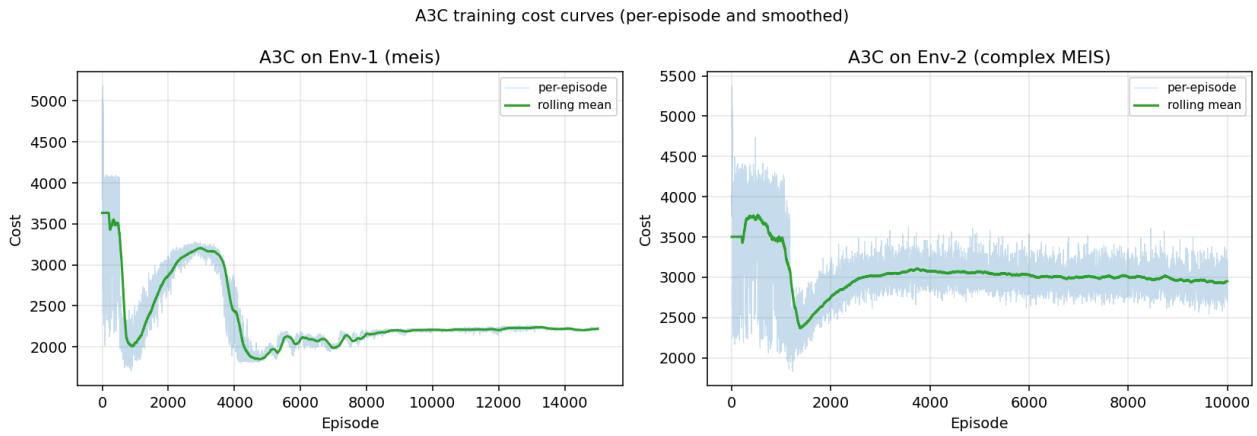

Figure 2: A3C training cost on Env-1 (left) and Env-2 (right). The Env-1 curve shows a mid-training bump around episodes 1500–4500 (policy / entropy oscillation) that ultimately resolves to a stable minimum. On Env-2, convergence settles on a higher cost plateau that, as Section 5.5 will show, is still worse than a retuned heuristic.

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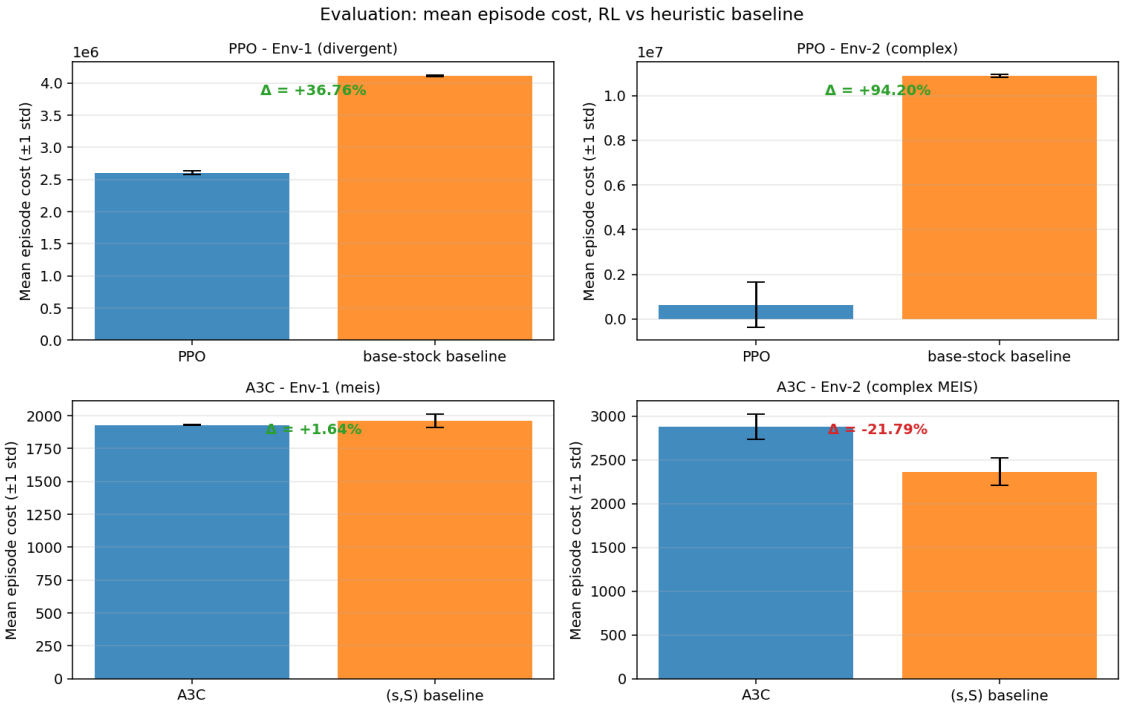

Figure 3: Mean episode cost (lower is better). Error bars are one standard deviation across evaluation episodes. Green = RL wins; red = RL loses. Absolute cost scales differ across algorithms (different cost coefficients and episode lengths); only the within-cell RL vs. baseline gap is interpretable.

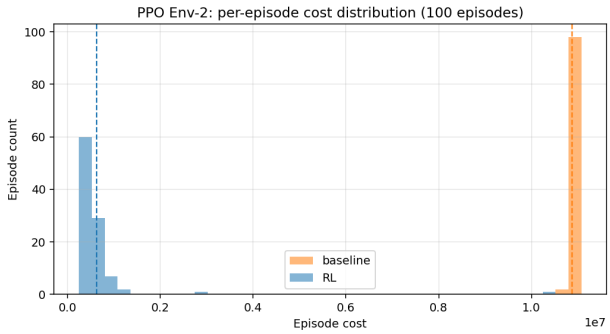

Figure 4: PPO Env-2 per-episode cost distribution (100 episodes). The bulk of PPO episodes (blue) concentrates well below the baseline mode (orange), but a clear right tail of episodes sit close to the baseline mode — these are shock episodes PPO has not learned to hedge.

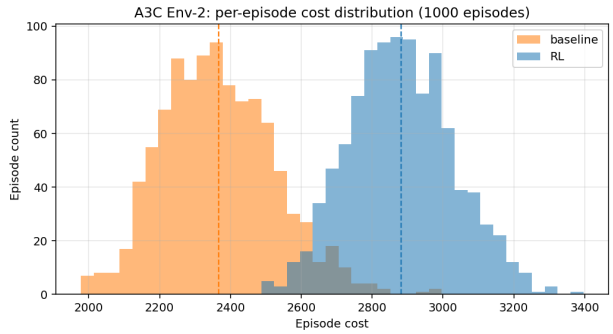

Figure 5: A3C Env-2 per-episode cost distribution (1000 episodes). The baseline (s, S) distribution (orange) lies almost entirely to the left of the A3C distribution (blue): at this training budget, A3C is stochastically dominated by the retuned heuristic.

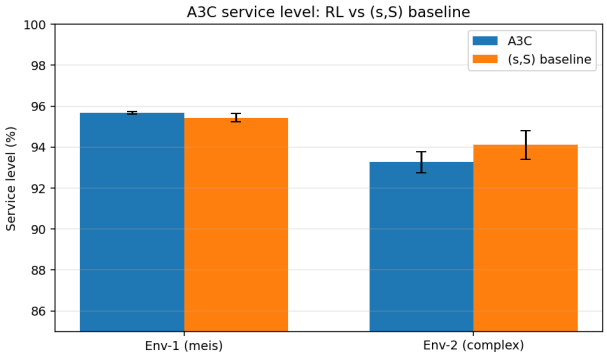

Figure 6: A3C service level on Env-1 and Env-2. On Env-2, A3C is worse than the retuned (s, S) on service level too, mirroring the cost result.